

Tutorial 8 report

EE5311 (Digital IC design)

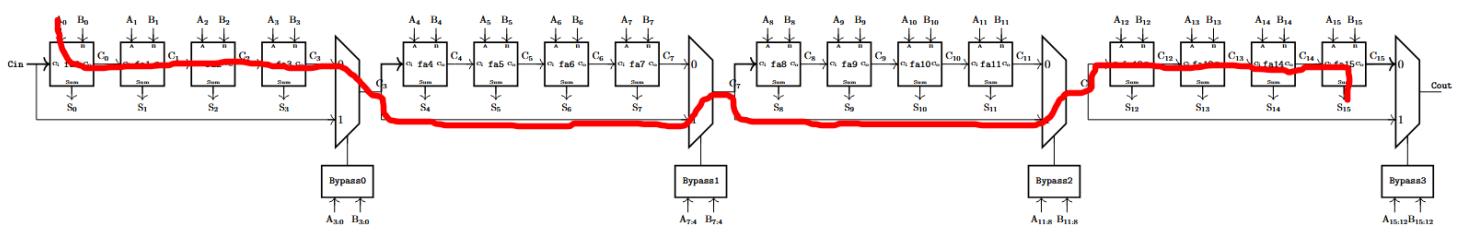
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Question 1

Part a)

- The critical path, starting from the input, goes through all the 4 full adders in the 1st stage (where the 1st full adder generates/kills the carry and the other 3 adders propagate). The path then passes through port 1 of the MUXs at the end of the 2nd and 3rd stages. After this, it propagates through the 1st 3 full adders of the last stage and ends at the sum of the last full adder, S15.

- Traced critical path on the 16-bit carry look ahead adder –



- An input combination that activates the critical path would be –

A = 1111111111111111

B = 0000000000000001

Cin = 0

- The critical path delay would be –

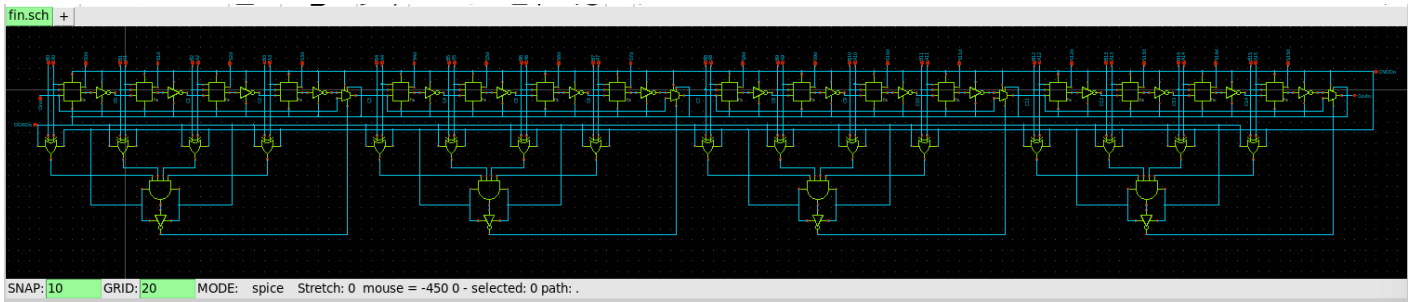
$$t_{\text{critical}} = 4 t_{\text{carry}} + 3 t_{\text{mux}} + 3 t_{\text{carry}} + t_{\text{sum}}$$

The first term $4 t_{\text{carry}}$ consists of $1 t_{\text{carry}}$ arising from the generate/kill delays of the 1st full adder and $3 t_{\text{carry}}$ from the propagation delay of the next 3 adders.

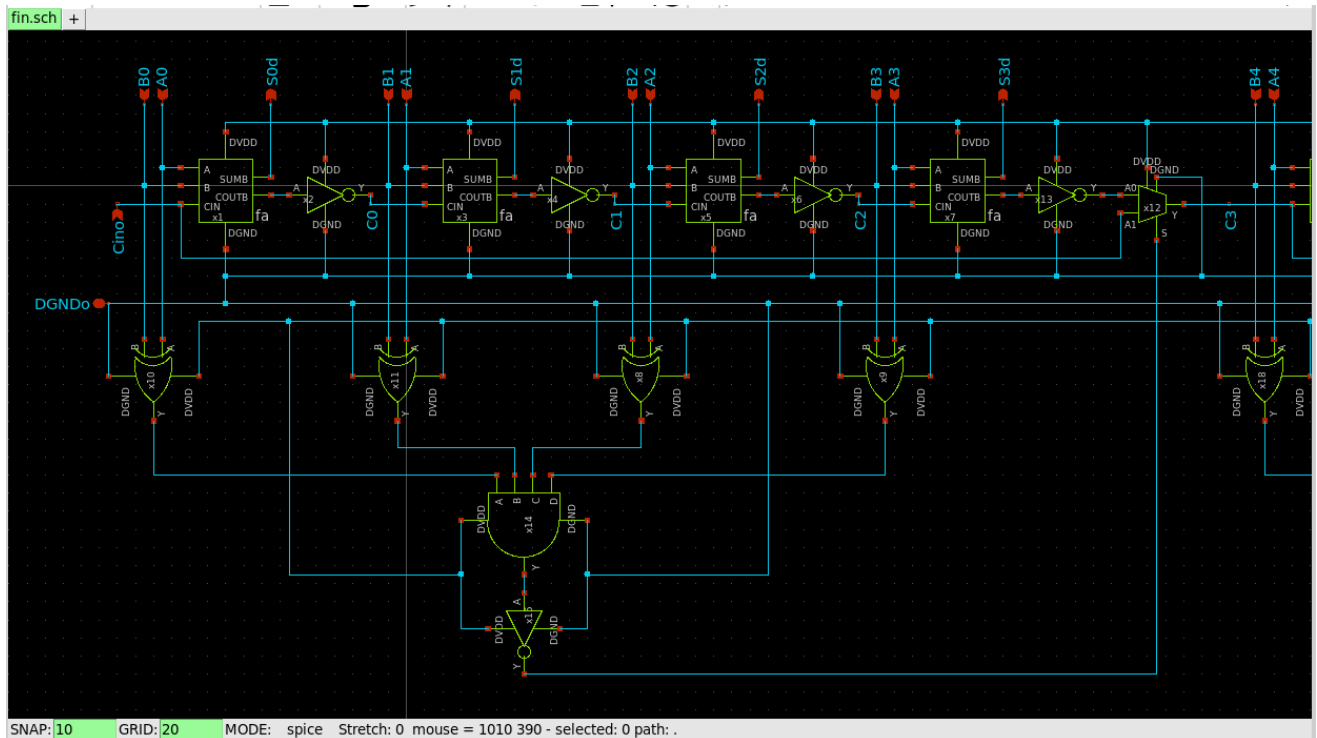
- Any other input combination, for example one that does not propagate through the 2nd and 3rd stages would be slower, since this means that a carry kill/generate would make the carry available immediately at that stage. This implies the availability of the output carry of the stage at the mux at at most $4 t_{\text{carry}}$ and that it would now have to travel through lesser MUXs to reach the final stage than if the entire stage had propagated.

Part b)

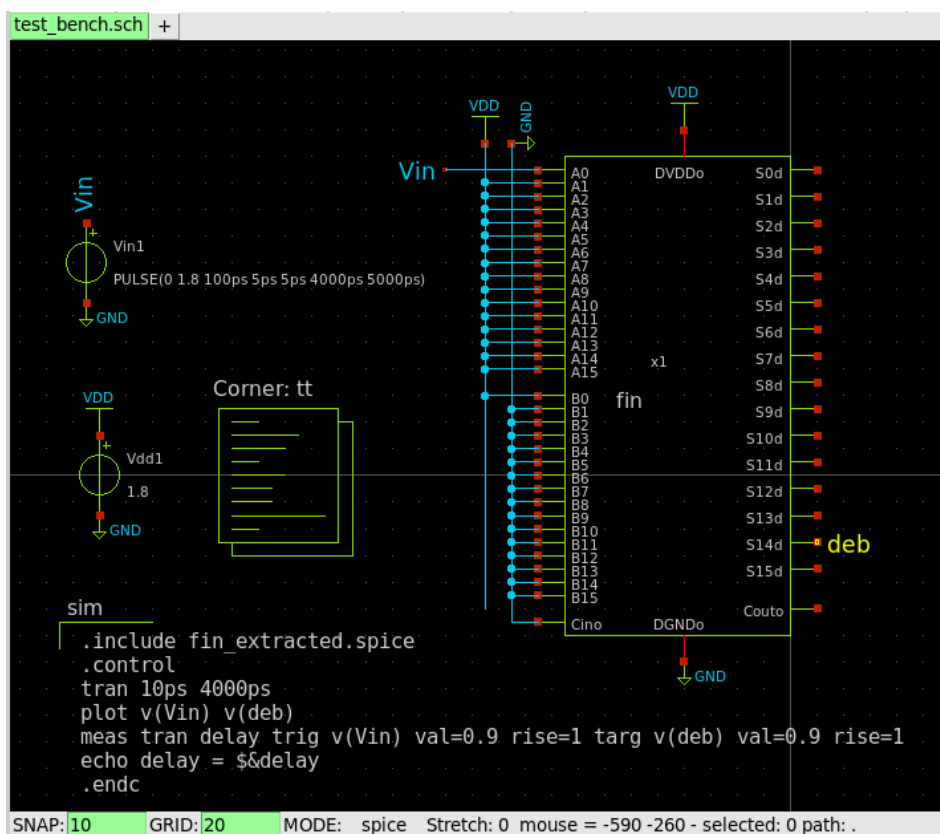
- The schematic used for the 16-bit carry look-ahead adder –



- A zoomed look at one of its 4 stages –

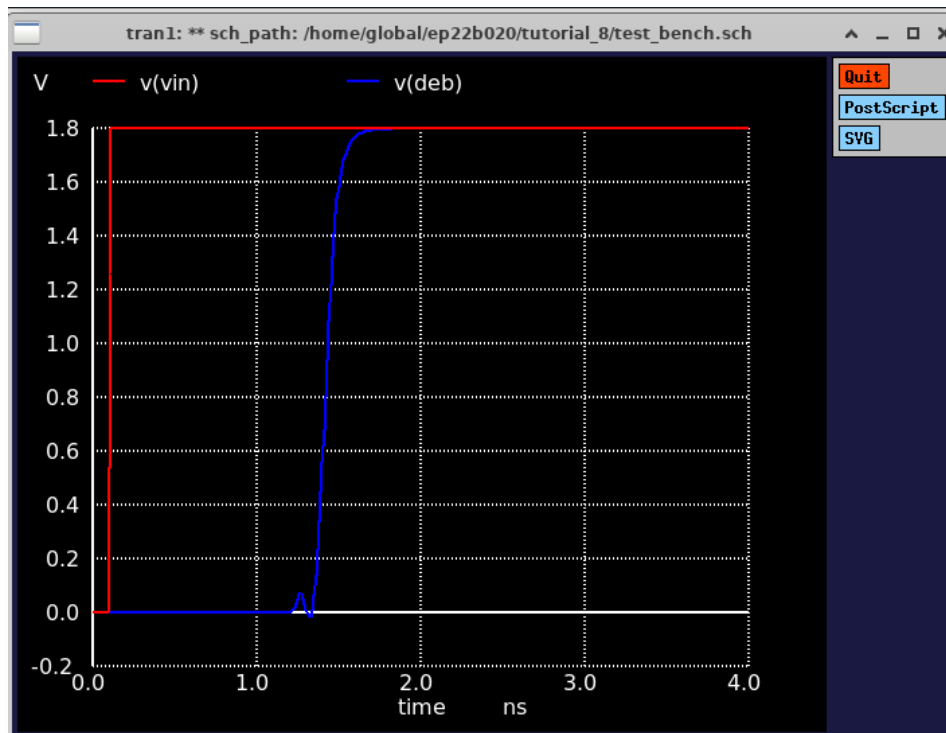


- xschem circuit and code used for measuring the critical delay –



- $t_{\text{critical}} = 1.322\text{ns}$

```
Reference value : 3.92740e-09
No. of Data Rows : 416
delay          = 1.322061e-09 targ= 1.424561e-09 trig= 1.025000e-10
delay = 1.32206E-09
ngspice 17 -> □
```



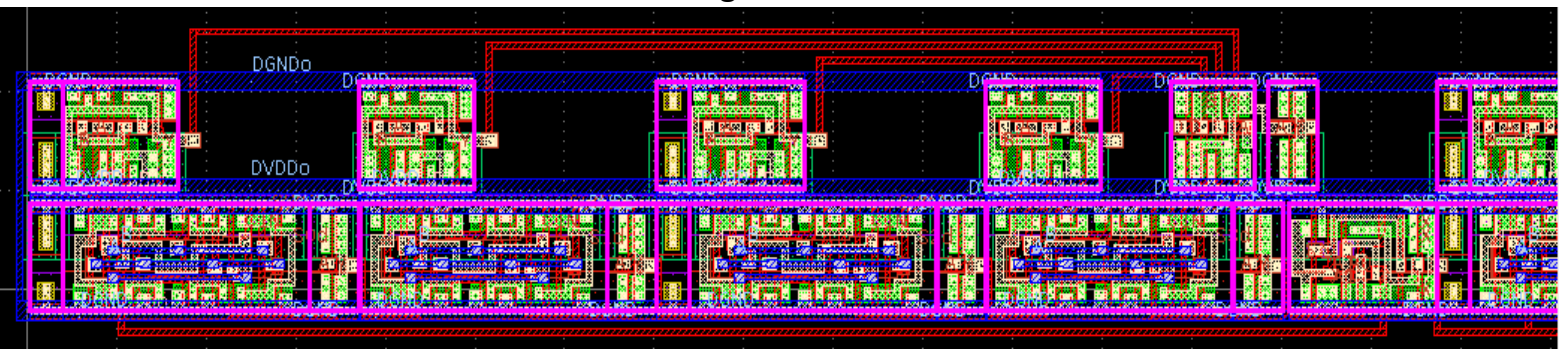
Question 2

Part a)

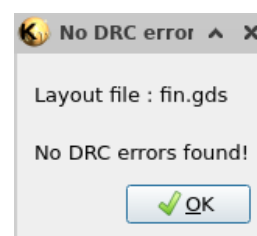
- Layout for the 16-bit carry look-ahead adder used –



- A zoomed look at one of its 4 stages –



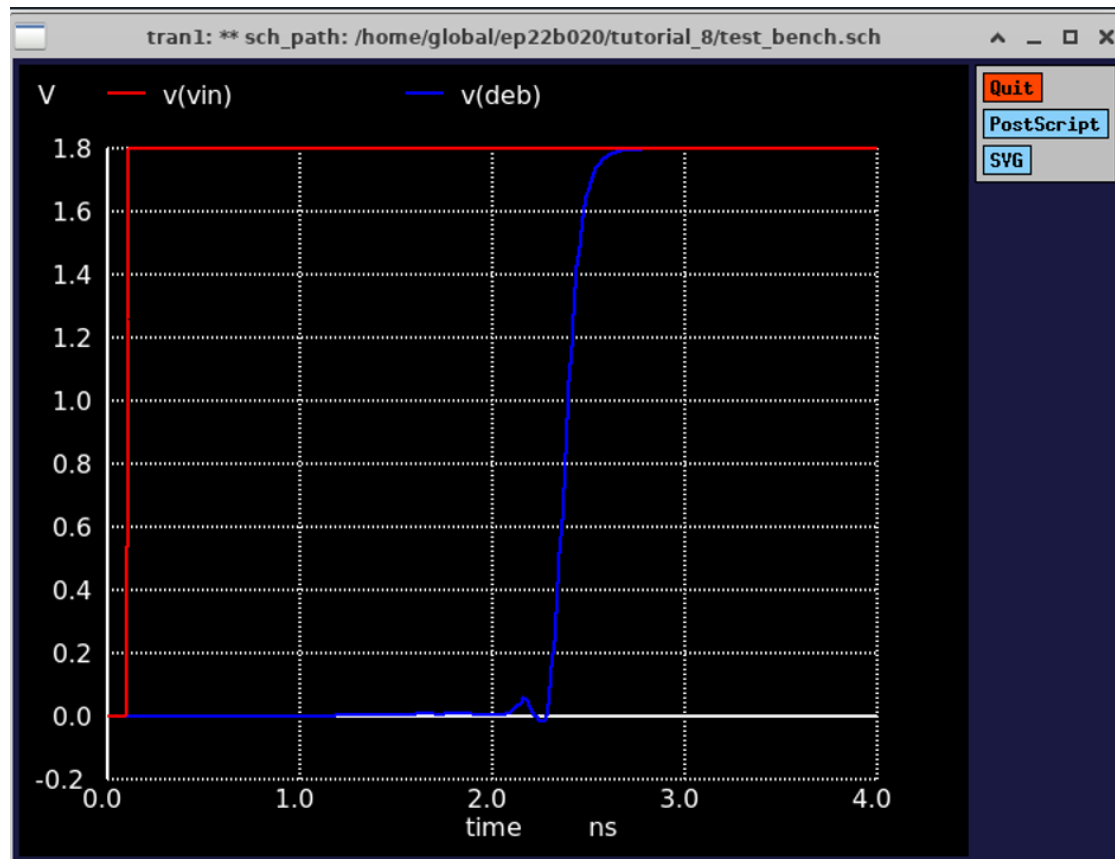
- The layout passed DRC and LVS, ensuring its compatibility with the schematic –



Part b)

- After extracting the layout parasitics into the schematic, $t_{\text{critical}} = 2.28\text{ns}$

```
Reference value : 4.00000e-09  
No. of Data Rows : 416  
delay          = 2.280308e-09 targ= 2.382808e-09 trig= 1.025000e-10  
delay = 2.28031E-09  
ngspice 17 -> □
```



- We see that after including the parasitics arising from interconnects, the delay indeed increases.